

APSC 171 – CALCULUS I

Course Syllabus – Fall 2026

This is your course syllabus. Please download the file and keep it for future reference.

TEACHING TEAM

LAND ACKNOWLEDGEMENT

Queen's University is situated on traditional Anishinaabe and Haudenosaunee Territory. See: <http://www.queensu.ca/encyclopedia/t/traditional-territories>

INCLUSIVITY STATEMENT

Queen's students, faculty, and staff come from every imaginable background – small towns and suburbs, urban high rises, Indigenous communities, and from more than 100 countries around the world. You belong here: <https://www.queensu.ca/vpcei/support/overview>.

COURSE INSTRUCTIONAL TEAM

Alan Ableson (he/him; MECH)

Peter Gilbert (CHEE)

Félix Parraud (Math & Stats)

Neige Paulet (Math & Stats)

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Office hours: TBA - see OnQ after first week of classes

APSC 171 (F 2.8-0-0.5 3.3)

COURSE DESCRIPTION

Graphs and derivatives of vector-valued functions; related applications. Implicit derivatives and related rate applications. Fundamental Theorem of Calculus, Riemann integral; applications to problems involving areas, volumes, mass, charge, work, etc. Integration by substitution, by parts, and partial fractions. Introduction to second-order differential equations and complex numbers.

(40/0/0/0/0) (Mathematics/Natural Sciences/Complementary Studies/Engineering Science/Engineering Design)

MODE OF INSTRUCTION

The delivery mode of this course is blended.

PRE-REQUISITE KNOWLEDGE

This course is designed for learners with Ontario Grade 12 Calculus and Vectors, or a similar high-school level background.

COURSE LEARNING OUTCOMES (CLO)

By the end of this course, students should be able to:

CLO	DESCRIPTION	INDICATOR ¹
CLO 1	Construct mathematical descriptions or expressions to model a real-world problem.	KB-Mathematics
CLO 2	Use an appropriate derivative-related method to solve a mathematical problem that arises from modeling a real-world problem.	KB-Mathematics
CLO 3	Use an appropriate integral-related method to solve a mathematical problem that arises from modeling a real-world problem.	KB-Mathematics
CLO 4	Use an appropriate differential equation solution technique to solve a mathematical problem that arises from modeling a real-world problem.	KB-Mathematics
CLO 5	Select and describe appropriate numerical methods to solve mathematical problems that arise from modeling a real-world problem.	KB-Mathematics
CLO 6	Use the solution to mathematical problems to inform the real-world problem that gave rise to it.	KB-Mathematics

¹ For further information on graduate attributes and indicators, please click [here](#).

COURSE EVALUATION

Assessment Weighting

Assessment Tool	Due Date	Weight		Alignment with CLOs
Completing QEng Prep Calculus Quiz	1 st Monday after start of classes	1% (bonus)	For students who get 70% or greater in their 1 st -3 rd attempts	
Weekly Assignment (Webwork)	Weekly on Thursdays, 3 PM	5% (best 10 of 12)		1-5
In-Class Participation (Clicker Questions) And Pre-Class Participation (Pre-Class Quizzes)	Clickers in Lectures; Pre-Class by 11:59 PM the night before your weekly Peer Instruction class	5% (full marks for 8 out of 11; starts Week 2)		
Test 1	Tue Oct 6 – 6:30 PM (Week 5)	22.5%	OR: Best Test 22.5% and Exam 67.5%	1-2, 6
Test 2	Tue Nov 10 – 6:30 PM (Week 9)	22.5%		1-3, 5, 6
Final Exam	Exam Period	45%		1-6
Enrichment Opportunities		Up to 1% (bonus)	See Information Pages/Enrichment in OnQ	
		100%		

Note 1: We will automatically compute two final marks for each student, either using both (Test 1 and Test 2), or just a student's best test, along with the final exam; see the [22.5%/22.5%/45%] vs [22.5%/67.5%] weights in the table. Each student's final grade will automatically be the **higher** of those two final marks.

Note 2: The expectation is that students will be able to demonstrate competency on all of the course material by having a passing average over all the proctored assessments;

students who would pass by the calculations in the Assessment Weighting table, but who have lower than 50% average on the 90 marks assigned for Test 1, Test 2 and Final Exam, will receive an FR final grade. In this case, students may enter the Extended Program to retake the course, or if they meet the eligibility criteria they may take the APSC 171 rewrite exam ("supplemental exam") in February. See:

<https://smithengineering.queensu.ca/first-year/rewrite-exams>

ASSESSMENT DESCRIPTIONS

Tests and Exams

Tests and Exams will require answers written out in full with all the steps explained. We are more interested in seeing the reasons for your answers and whether you understand the concepts involved than in the numerical answer. Also, if you explain your solutions clearly, we can give part marks where appropriate.

We allow tests and exams to be written in either pencil or pen. No resources are allowed during exams other than your calculator (i.e., no notes, formulas, textbooks, etc.). See the note on Academic Integrity below.

Difficulty: The test and exam questions will be as hard as some of the harder practice problems. On a test or exam, there will not be many marks for merely being able to do routine calculations and procedures - we will take it for granted that you can do those.

Previous Tests and Exams: Looking at APSC 171 examinations from past years will help you to be prepared for the style of the questions we will ask. See the course OnQ site for old tests and exams, or the central Exam Bank for old exams (<https://exambank.library.queensu.ca/>).

Missed tests: (with appropriate notification) will not be re-taken; if a student must miss a test, we will use the 22.5% Best Test/67.5% Exam mark distribution.

Weekly Assignments (Webwork)

There will be weekly assignments to be submitted online using the Webwork platform. A link to Webwork can be found from the course OnQ page.

Each student will get the same style of questions, but parts of each question will be randomized. The software marks your submissions right away and you may repeatedly attempt the same assignment until you get it correct. You will only be able to access each weekly assignment up until its matching deadline on the Thursday after the week the material was covered. The best 10 out of 12 Webwork week grades will be counted.

In-Class Participation

We will be using a software-based clicker solution during the lectures. Responses will be accumulated over each week, and students with 8 or more weeks of participation will receive the full grade. Fewer weeks participation will be pro-rated as a fraction of 8 weeks.

Note: In-class participation will not be counted for Week 1, due to the Labour Day holiday shortening the week for some sections.

Pre-Class Quizzes

Each week, the night before your second class meeting of the week, an OnQ quiz will be due. This quiz will be short (3-4 questions), and is meant to give practice with the material covered in video lectures, before coming to class. The grade on the quiz will not matter as it is meant as a learning tool. Instead, we will count the number of weeks each student completes the quiz: students who complete 8 or more out of the 11 quizzes will receive full marks. Fewer weeks completion of the quizzes will be pro-rated as a fraction of 8 weeks.

Note: Pre-class Quizzes will only start in Week 2, due to the Labour Day holiday shortening the week for some sections.

GRADING

All assessments in this course will receive numerical percentage marks. The final grade you receive for the course will be derived by converting your numerical course average to a letter grade according to the established [Grade Point Index](#).

Feedback on Assessments

The teaching team will provide feedback on graded activities.

- For the Weekly Assignments (Webwork) and Pre-Class Quizzes, the feedback will be immediate.
- For the written assessments (tests and final exam) you can expect your grade and feedback within fourteen days of the test date.

Accessing Your Final Grade

Your final grades will show on SOLUS. Official transcripts showing final grades will be available on the Official Grade Release Date. Please note that in official transcripts, a mark of IN (incomplete) is considered a grade, and your transcript is released with this grade.

COURSE MATERIALS

OPTIONAL Textbook

Calculus: Early Transcendentals by James Stewart. Any edition is acceptable. New; \$225; used from \$60.

Other Material

- In-class participation will be done using Qlicker at <https://qlicker.queensu.ca> . Accounts are free; login using the "Login/Create Account" button.
- Webwork for weekly homework is free for students. Accounts will be automatically created for each student.
- All other course material is accessible via OnQ.

Required Calculator

A Casio 991-series calculator is required. **ONLY** this type of non-programmable, non-communicating calculator will be allowed during tests and exams.

Suggested Time Commitment

This course represents a study period of one semester spanning 12 weeks. Learners can expect to invest on average 9-11 hours per week in this course.

WEEKLY COURSE LEARNING OUTCOMES

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Wee k	Learning Outcomes	Assessment
1	Derivatives and Vectors By the end of this week, learners will be able to: • Evaluate derivatives of all common functions.[CLO2] • Compute derivatives of more complicated functions using the product, quotient and chain rules. [CLO2] • Compute and interpret sums of vectors, and scalar and vector products of vectors.[CLO1]	
2	Vector-Valued Functions By the end of this week, learners will be able to: • Define vector-valued functions (parametric curves). [CLO1] • Sketch vector-valued functions. [CLO1] • Determine if vector-valued functions collide or intersect. [CLO1]	
3	Derivatives with Vector-Valued Functions By the end of this week, learners will be able to: • Compute and use velocity in parametric curves. [CLO2] • Compute and use acceleration in parametric curves. [CLO2]	

Week	Learning Outcomes	Assessment
4	Inverse Trig, Tangent Lines and Linearization By the end of this week, learners will be able to: • Compute and interpret results with inverse trigonometric functions and their derivatives. [CLO1] • Create and use linearization/tangent line formulas. [CLO2]	
5	Implicit Derivatives, Related Rates and Optimization By the end of this week, learners will be able to: • Compute implicit derivatives. [CLO2] • Solve problems involving related rates. [CLO2] • Apply the first derivative test and the second derivative test for identifying local maxima and minima. [CLO2] • Distinguish global vs. local extrema. [CLO2] • Solve optimization word problems. [CLO6]	Test 1 [CLO1][CLO2][CLO6]
6	Defining and Estimating Integrals as Areas By the end of this week, learners will be able to: • Determine how to calculate the area described by a function. [CLO4] • Estimate the area defined by a function using numerical methods. [CLO4] • Define the definite integral. [CLO3] • Connect the relationship between the definite integral and area above and below the axis. [CLO3]	
7	The Fundamental Theorem of Calculus By the end of this week, learners will be able to: • List properties of definite integrals. [CLO3] • State the Fundamental Theorem of Calculus. [CLO3] • Compute simple anti-derivatives and definite integrals. [CLO3]	

Week	Learning Outcomes	Assessment
8	Integration by Parts and Substitution	
	By the end of this week, learners will be able to: • Apply the guess-and-check method for anti-differentiation. [CLO3] • Demonstrate the substitution method for anti-differentiation. [CLO3] • Demonstrate integration by parts. [CLO3] • Apply a slicing approach to construct integrals for areas. [CLO3]	
9	Integrals for Volume and Work; Partial Fractions	Test 2 [CLO1][CLO3] [CLO5][CLO6]
	By the end of this week, learners will be able to: • Apply slicing approaches to construct integrals for volumes and work. [CLO6] • Evaluating integrals using partial fractions. [CLO3]	
	Differential Equations and Complex Numbers 1	
10	By the end of this week, learners will be able to: • Represent a spring/mass system with differential equations [CLO4] • Present harmonic motion in mathematical form. [CLO1] • Define and use polar and Cartesian forms of Complex Numbers. [CLO1]	
11	Differential Equations and Complex Numbers 2	
	By the end of this week, learners will be able to: • State Euler's Formula. [CLO1] • Use the complex exponential function to convert between Cartesian and polar forms. [CLO1] • Solve for the behaviours of a spring/mass system using complex numbers. [CLO4]	

Week	Learning Outcomes	Assessment
12	Review By the end of this week, learners will be able to: • Summarize and perform tasks from the earlier work from the course.	Leads to Final Exam [CLO1][CLO2][CLO3] [CLO4][CLO5][CLO6]

COURSE COMMUNICATION

QUESTIONS ABOUT COURSE MATERIAL

Questions or comments regarding the course material that can be of benefit to other students should be posted in the Q&A forum on the class website. The instructor, TAs, and students are encouraged to answer these questions directly in the discussion forum for the benefit of everyone in the course.

COURSE ANNOUNCEMENTS

The instructor will routinely post course news in the Announcements section on the main course homepage on OnQ. Please sign up to be automatically notified by email when the instructor posts new information in the Announcements section. Instructions on how to modify your notifications are found in the **Begin Here** section of the onQ course site.

OFFICE HOURS

In addition to interaction in the Q&A discussion forums, you will have the opportunity to interact with either a TA or the instructor during office hours. The instructor will provide a schedule of availability at the beginning of the term.

CONFIDENTIAL MATTERS

If you have a confidential matter you would like to discuss with your instructor, their contact details are on the first page of this document. Expect email replies within 48 hours.

ABSENCES (ACADEMIC CONSIDERATIONS) AND MISSED ASSIGNMENTS

Students who miss assignments, quizzes, classes, labs, tutorials or other course deliverables due to unforeseen personal circumstances outside of their control can apply for Academic Consideration. For information on academic considerations due to extenuating circumstances, please review the information on the [Smith Engineering website](#). Note that unacceptable reasons include extracurricular activities, work, travel plans, scheduling conflicts, generally being behind on schoolwork, etc. Do not schedule travel during midterms and final exams, as travel and work are not acceptable reasons for granting academic considerations.

For any communication around academic considerations, please use the course email address,
apsc171@queensu.ca

LATE SUBMISSION POLICY

In APSC 171, we do not accept late submissions for the Weekly Assignments or Pre- and In-Class Participation activities. Instead, we allow students to drop up to 2 Weekly Assignments, and up to 3 missed Pre- and In-Class Participation activities to allow for the occasional absence or forgetful moment.

STANDARD QUEEN'S AND SMITH ENGINEERING POLICIES

NETIQUETTE

In this course, you may be expected to communicate with your peers and the teaching team through electronic communication. You are expected to use the utmost respect in your dealings with your colleagues or when participating in activities, discussions, and online communication.

The following is a list of netiquette guidelines. Please read them carefully and use them to guide your online communication in this course and beyond.

1. Make a personal commitment to learn about, understand, and support your peers.
2. Assume the best of others and expect the best of them.
3. Acknowledge the impact of oppression on the lives of other people and make sure your writing is respectful and inclusive.
4. Recognize and value the experiences, abilities, and knowledge each person brings.
5. Pay close attention to what your peers write before you respond. Think through and re-read your writings before you post or send them to others.
6. It's alright to disagree with ideas, but do not make personal attacks.
7. Be open to be challenged or confronted on your ideas and challenge others with the intent of facilitating growth. Do not demean or embarrass others.
8. Encourage others to develop and share their ideas.

STUDENT CODE OF CONDUCT

Queen's University values maintaining an environment free of, and will not tolerate, harassment, discrimination, and reprisal. The Student Code of Conduct applies to all students at Queen's. It outlines the activities and behaviours that could be considered Non-Academic Misconduct (NAM). The Code also describes the NAM process and the sanctions that could be imposed on a student found responsible for a violation. All students should be familiar with the Student code of conduct and related policies on sexual violence prevention and response and harassment and discrimination prevention and response.

<https://www.queensu.ca/nonacademicmisconduct/policies>

COPYRIGHT

Course materials created by the course instructor, including all slides, presentations, synchronous and asynchronous course recordings, handouts, tests, exams, and other similar course materials, are the intellectual property of the instructor. It is a departure from academic integrity to distribute, publicly post, sell or otherwise disseminate an instructor's course materials or to provide an instructor's course materials to anyone else for distribution, posting, sale or other means of dissemination, without the instructor's **express**

consent. A student who engages in such conduct may be subject to penalty for a departure from academic integrity and may also face adverse legal consequences for infringement of intellectual property rights and, with respect to recordings, potentially privacy violations of other students.

ACADEMIC INTEGRITY

As an engineering student, you have made a decision to join us in the profession of engineering, a long-respected profession with high standards of behaviour. As future engineers, we expect you to behave with integrity at all times. Please note that Engineers have a duty to:

- Act at all times with devotion to the high ideals of personal honour and professional integrity.
- Give proper credit for engineering work

The standard of behaviour expected of professional engineers is explained in the [Professional Engineers Ontario Code of Ethics](#). Information on policies concerning academic integrity is available in the [Queen's University Code of Conduct](#), in the [Senate Academic Integrity Policy Statement](#), on the [Smith Engineering website](#), and from your instructor.

Departures from academic integrity include plagiarism, use of unauthorized materials or services, facilitation, forgery, falsification, unauthorized use of intellectual property, and collaboration, and are antithetical to the development of an academic community at Queen's. Given the seriousness of these matters, actions which contravene the regulation on academic integrity carry sanctions that can range from a warning or the loss of grades on an assignment to the failure of a course to a requirement to withdraw from the University.

In the case of online or remotely proctored exams, impersonating another student, copying from another student, making information available to another student about the exam questions or possible answers, posting materials to online services, communicating with another person during an exam or about an exam during the exam window, or accessing unauthorized materials, including internet sources and using unauthorized materials, including smart devices, are actions in contravention of academic integrity.

PLAGIARISM/USE OF GENERATIVE ARTIFICIAL INTELLIGENCE (GENAI) TOOLS:

Students must submit their own work and cite the work that is not theirs. Any other use constitutes a Departure from Academic Integrity.

GenAI tools are usage:

1. GenAI tools are permissible in relation to any self-study part of the course.
2. Gen AI tools are **not** allowed during the on-paper midterms and final exam.

INVALID EXAMS

An exam may be declared invalid in case of an interruption in an in-person examination; if the instructions in a remote or online exam were not followed; if the student uploads wrong materials; or if a situation arises where the integrity of the exam cannot be verified. If an exam is declared invalid, the student may be granted a re-write.

ACADEMIC AND STUDENT SUPPORT

Queen's has a robust set of supports available to you, including the [Library](#), [Student Academic Success Services \(Learning Strategies and Writing Centre\)](#), and [Career Services](#). Learners are encouraged to visit the Smith Engineering [Current Students](#) web portal for information about various other policies such as academic advisors, registration, student exchanges, awards and scholarships, etc. Students are also encouraged to review the information that is available in the EngQ Hub, posted in onQ.

ABSENCES - ACADEMIC CONSIDERATIONS

Academic Consideration is a process by which students request validated time away from their academics when they are experiencing an extenuating circumstance or personal circumstances beyond a student's control that interfere with their ability to complete academic requirements.

For additional information and to apply for academic consideration, please review the information on the [Smith Engineering's Academic Consideration \(Absences\) website](#).

ACADEMIC ACCOMMODATIONS

Academic Accommodations are available when students need an adjustment to their learning environment due to a reason related to a Human Rights protected ground. Below there is information on the most common Academic Accommodations requested by students. For more information, please see [Smith Engineering's Academic Accommodation website](#).

ABSENCES – REQUESTS FOR EXCUSED ABSENCES FOR SIGNIFICANT EVENTS (REASE)

Students participating in a Varsity Athletics event, the student reserve forces, a non-varsity event at the provincial, national, or international level, or an event to which you were invited as a distinguished guest can submit a Request for Excused Absence for Significant Event following the REASE process outlined on the Student Affairs webpage.

Students participating in Design Competitions supported by Smith Engineering can submit a Request for Excused Absence for Significant Events (REASE) following the process outlined on the [Smith Engineering's Academic Consideration \(Absences\) website](#).

All Requests for Excused Absences for Significant Events must be submitted to the Faculty a minimum of 2 weeks prior to the event start date.

RELIGIOUS ACCOMMODATIONS

Students in need of accommodation for religious observance are asked to complete the [Religious Observance and Academics Activities](#) form within a week of receiving their course syllabus (each term) and submit the form via email to the Engineering Program Advisor, (Accommodations & Considerations), engineering.aac@queensu.ca

- Alternative assignments are considered a “reasonable accommodation” under the Ontario Human Rights Code
- Please refer to the [Queen’s Multi-faith Calendar](#) for all of this year’s faith dates
- Students with questions about their rights and responsibilities regarding religious accommodations can contact chaplain@queensu.ca

OTHER HUMAN-RIGHTS BASED ACCOMMODATIONS

Engineering students who need accommodation on Human Rights grounds are asked to contact, in confidence, the Engineering Program Advisor, (Accommodations & Considerations) engineering.aac@queensu.ca or to call 613-533-6000 x 78013.

ACCOMMODATIONS FOR DISABILITIES

Queen's University is committed to working with students with disabilities to remove barriers to their academic goals. Queen's Student Accessibility Services (QSAS), students with disabilities, instructors, and faculty staff work together to provide and implement academic accommodations designed to allow students with disabilities equitable access to all course material (including in-class as well as exams). If you are a student currently experiencing barriers to your academics due to disability related reasons, and you would like to understand whether academic accommodations could support the removal of those barriers, please visit the [QSAS website](#) to learn more about academic accommodation.

To start the registration process with QSAS, click the “**Access Ventus**” button found on the [Ventus student portal](#).

Ventus is an online portal that connects students, instructors, Queen's Student Accessibility Services, the Exam's Office, and other support services in the process to request, assess, and implement academic accommodations. To learn more about Ventus, visit A Visual Guide to Ventus for Students:

<https://www.queensu.ca/ventus-support/students/visual-guide-ventus-students>

For questions or assistance with requesting Academic Consideration or Accommodation, contact the Smith Engineering Program Advisor (Accommodations and Considerations) at engineering.aac@queensu.ca

Every effort has been made to provide course materials that are accessible. For further information on accessibility compliance of the educational technologies used in this course, please consult the links below.

STUDENT WELLBEING

Success in Engineering includes taking care of your wellbeing. If you're experiencing stress, anxiety, burnout, personal challenges, or you're simply not sure where to turn, [EngWell](#) is here to help.

EngWell provides confidential support through our [Wellness Advisor](#), [Embedded Counsellors](#), [mental health programming](#), and referrals to academic and campus resources. Whether you need short term guidance, help navigating university supports, or someone to talk to, [we encourage you to reach out early](#).

EDUCATIONAL TECHNOLOGY

Educational Technology	Accessibility Compliance Information
onQ (Brightspace Learning Management System by D2L)	https://www.d2l.com/accessibility/standards/
Qlicker In-Class Response System	https://qlicker.queensu.ca/

If you find any element of this course difficult to access, please discuss with your instructor how you can obtain an accommodation.

TECHNICAL SUPPORT

Some basic comfort levels with basic hardware and software skills are required for this course. If you require technical assistance, please contact [Technical Support](#).